

CSE4502 Operating Systems Lab 2B

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1 Part A: File Type Identifier (Case and Conditionals)

1.1 Problem Statement

Create a Bash script named `file_type.sh` that determines the type of a file based on its extension (e.g., `.txt`, `.sh`, `.c`, `.jpg`).

1.2 Logic

1. Accept a filename as a command-line argument.
2. Use a `case` statement to identify the file type.
3. Print the result based on the extension.

1.3 Hints

- Extract the file extension using parameter expansion:
`ext="${filename##*.}"`.
- Convert extension to lowercase for case-insensitive matching using :
`ext_lower=$(echo "$ext" tr '[:upper:]' '[:lower:]') |`

1.4 Example Output

```
$ ./file_type.sh photo.jpg
File type: Image File
```

```
$ ./file_type.sh report.txt
File type: Text File
```

```
$ ./file_type.sh main.c
File type: C Source File
```

2 Part B: Sequential Character Printing Using Semaphores

2.1 Problem Statement

Given an integer n such that $1 \leq n \leq 26$, create n concurrent threads. Each thread is responsible for printing exactly one uppercase character from the English alphabet. Despite starting in an arbitrary order, the threads must print characters sequentially in alphabetical order.

2.2 Thread Responsibilities

- Thread 1 prints character A
- Thread 2 prints character B
- $\vdots$
- Thread n prints character corresponding to ('A' + n - 1)

Each thread prints exactly one character and then terminates.

2.3 Synchronization Constraints

- Threads may start execution in any order.
- Output must always appear in strictly increasing alphabetical order.
- Each thread waits a random amount of time before printing, The execution of the threads should be sequential even after applying random delays. Use the following snippet to introduce delays:

```
void delay_ms() {  
    unsigned int milliseconds = (rand() * 100)%500;  
    usleep(milliseconds * 1000); // Convert ms to microseconds  
}
```

2.4 Expected Output

For a given value of n , the output must be:

A B C ...

ending at the n -th character.

2.5 Example

For $n = 4$, four threads are created and may start in any order, but the output must be:

A B C D